

Practical No. 9

Aim: To create and configure a cloud-based database server using Amazon RDS and connect it with applications in the AWS environment.

Theory:

Amazon Relational Database Service (RDS) is a managed cloud database service that simplifies the setup, operation, and scaling of relational databases. It supports multiple database engines including MySQL, PostgreSQL, Oracle, and SQL Server, and handles routine tasks such as backups, patching, and replication automatically.

Amazon RDS supports Multi-AZ deployments, where a primary database instance is automatically replicated to a standby instance in a different Availability Zone. This ensures high availability and data durability in case of infrastructure failure.

Key Features of Amazon RDS

- Managed service — AWS handles backups, patching, and maintenance
- Multi-AZ support for high availability and automatic failover
- Supports multiple database engines including MySQL, PostgreSQL, and Oracle
- Easily scalable storage and compute resources

Advantages of Amazon RDS

- Reduces administrative overhead of managing a database server
- High availability through automated Multi-AZ replication
- Secure access control using VPC, security groups, and IAM

Steps:

Task 1: Create a Security Group for the RDS Instance

1. In the AWS Console, navigate to VPC and open Security groups. Click Create security group.
2. Set Security group name to DB Security Group, Description to Permit access from Web Security Group, and select Lab VPC.
3. Add an inbound rule with Type: MySQL/Aurora (port 3306) and Source: Web Security Group. Click Create security group.

Basic details
Security group name Info
DB Security Group
Name cannot be edited after creation.
Description Info
Permit access from Web Security Group
VPC Info
vpc-06e24a93cc292521f (Lab VPC)

Inbound rules Info			
Type Info	Protocol Info	Port range Info	Source Info
MySQL/Aurora	TCP	3306	Custom
sg-0cb2e13750295d6a			

Task 2: Create a DB Subnet Group

1. Navigate to RDS and choose Subnet groups. Click Create DB Subnet Group and set Name to DB-Subnet-Group, Description to DB Subnet Group, and VPC to Lab VPC.
2. Under Add Subnets, select Availability Zones us-east-1a and us-east-1b, and select subnets with CIDR ranges 10.0.1.0/24 and 10.0.3.0/24. Click Create.

Subnet group details
Name
You won't be able to modify the name after your
DB-Subnet-Group
Must contain from 1 to 255 characters. Alphanu
Description
DB Subnet Group
VPC
Choose a VPC identifier that corresponds to the s
Lab VPC (vpc-06e24a93cc292521f)
4 Subnets, 2 Availability Zones

Add subnets

Availability Zones
Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

us-east-1a X us-east-1b X

Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

Private Subnet 1
Subnet ID: subnet-05380a18b9f19f568 CIDR: 10.0.1.0/24 X

Private Subnet 2
Subnet ID: subnet-0291112494671ede1 CIDR: 10.0.3.0/24 X

Task 3: Create an Amazon RDS DB Instance

1. In RDS, choose Databases and click Create database. Select MySQL as the engine, Dev/Test as the template, and Multi-AZ DB instance for availability.
2. Under Settings, set DB instance identifier to lab-db, Master username to main, and Master password to lab-password.

DB instance identifier [Info](#)
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

lab-db

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter.

Credentials Settings

Master username [Info](#)
Type a login ID for the master user of your DB instance.

main

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - *most secure*
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed
Create your own password or have Amazon RDS generate one for you.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength [Average](#)

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' * @

Confirm master password [Info](#)

3. Set instance class to db.t3.micro, storage type to General Purpose SSD with 20 GB, and VPC to Lab VPC. Select DB Security Group and deselect default.

Instance type

db.t3.micro

2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

Storage

Storage type [Info](#)
Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp3)
Performance scales independently from storage

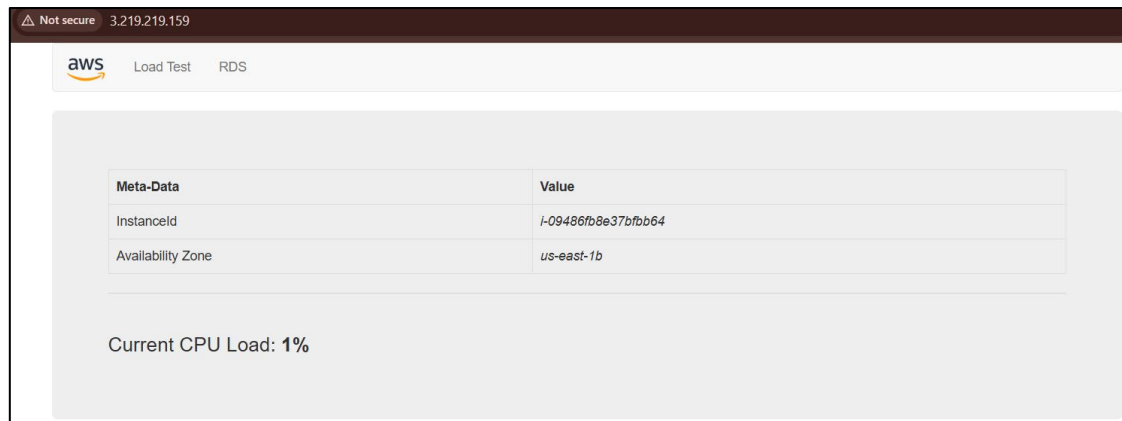
Allocated storage [Info](#)

20

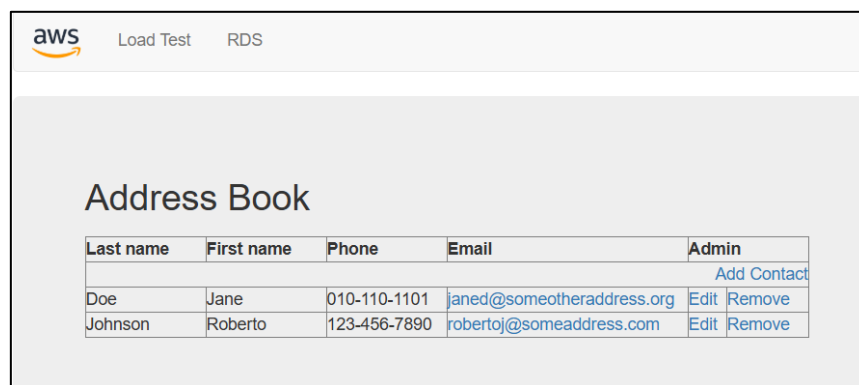
4. Under Additional configuration, set Initial database name to lab, and uncheck Enable automatic backups, Enhanced monitoring, and Enable encryption. Click Create database.
5. Wait for the database status to show Available, then copy the Endpoint value from the Connectivity & security section.

Task 4: Connect the Application to the Database

1. Open the WebServer IP address in a browser. The web application is displayed showing EC2 instance information.



- Click the RDS link at the top of the page. Enter the RDS Endpoint, Database: lab, Username: main, and Password: lab-password. Click Submit.
- The Address Book application loads, confirming the web application is successfully connected to the RDS database. Test by adding, editing, and removing contacts.



Conclusion: Thus, we have successfully created and configured a cloud-based database server using Amazon RDS and connected it with an application in the AWS environment.